

Himel Mallick, PhD, FASA

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ADDRESS Weill Cornell Medicine
425 E 61st Street
New York, NY 10065
E-mail: him4004@med.cornell.edu

Web:// himelmallick.org
GitHub:// [himelmallick](https://github.com/himelmallick)
Google Scholar:// [Himel Mallick](https://scholar.google.com/citations?user=HimelMallick)
LinkedIn:// [mallickhimel](https://www.linkedin.com/in/mallickhimel)

SUMMARY Data Scientist and Computational Biologist with two decades of experience in Statistics, Biostatistics, and AI/ML in academia and industry. Highly cited publication track record with over 50+ publications and widely used computational tools (10+ R packages). Specialized in Bayesian statistics, machine learning, and omics data science methodologies including multimodal integration, microbiome, single-cell, spatial transcriptomics, imaging, and digital pathology. Recipient of the IISA ECASDS Award, 2022 • 2023 Fellow of the ASA (FASA) • 2023 Elected Member of the ISI.

EDUCATION AND TRAINING **Postdoctoral Fellowship**, Biostatistics, Harvard University and Broad Institute 2019
Ph.D., Biostatistics, University of Alabama at Birmingham (GPA: 4.0/4.0) 2015
M.Sc., Statistics, IIT Kanpur (Class Rank: 4) 2009
B.Sc., Statistics, University of Calcutta (First Class Honours) 2007

EXPERIENCE **Tenure-track Faculty of Biostatistics, Population Health Sciences** 04/2023 – Present
Adjunct Faculty of Statistics and Data Science 06/2023 – Present
Adjunct Faculty of Gastroenterology and Hepatology 11/2025 – Present
Cornell University

- As a tenure-track faculty member at Cornell, I am building a cross-disciplinary team of statisticians, computational biologists, and AI scientists, bringing together powerful techniques from AI/ML, statistics, systems biology, and omics data science to develop reverse translation solutions for target identification and biomarker discovery.
- I hold a joint appointment in the Division of Gastroenterology and Hepatology as well as the Department of Statistics and Data Science, where my primary goal is to mentor and collaborate with Cornell investigators.

Senior/Associate Principal Scientist, Biostatistics 03/2019 – 04/2023
Merck Research Laboratories

- Led multiple data science projects providing end-to-end bioinformatics and biostatistics support across all stages of biomarker discovery and development including assay quality control, drafting SAP, and performing statistical analyses
- Supported Merck's internal efforts in the single-cell and spatial transcriptomics initiatives for translational oncology studies
- Liaised with digital pathology scientists and bioinformaticians on crucial statistical analyses for both exploratory and clinical development purposes
- Employed machine learning classification to the discovery of clinically actionable biomarkers from integrated multi-omics profiles to enable better disease outcome prediction and patient stratification
- Secured funding to initiate and execute multiple academic collaborations to conduct research on various 5-year strategic initiatives specified by the organization
- Successfully hired and co-mentored 4 Ph.D. summer interns towards the development and implementation of sophisticated statistical and AI/ML approaches for enabling personalized medicine
- Published research across several disease areas (infectious diseases, oncology, and microbiology) that utilized innovative statistical methodology and systems biology techniques spanning a wide range of translational applications

Postdoctoral Researcher, Computational Biology and Bioinformatics 10/2015 – 03/2019
Huttenhower lab, Harvard University and Broad Institute

- Lead developer of [MaAsLin 2](#), a R/Bioconductor package for associating microbial multi-omic data with arbitrarily complex clinical metadata (> 50,000 official downloads)
- Lead developer of [MelonnPan](#), a tool to predict metabolite profiles from metagenomic sequencing data using concepts from machine learning and ecology, implemented in R (> 250 citations)
- Contributed to grant writing, manuscript preparation, interdisciplinary collaborations, and teaching and mentoring of graduate and undergraduate students and trainees

Intern, Biostatistics Mayo Clinic	Summer 2015
Developed Bayesian adaptive trial designs for clinical trials utilizing surrogate endpoints in the presence of biomarkers	
Research Assistant, Biostatistics University of Alabama at Birmingham (UAB)	08/2010 – 04/2015
- Developed Bayesian machine learning methods for high-dimensional feature selection in personalized medicine applications - Developed and validated risk prediction models for assessing short-term mortality in obese adults - Conducted high-dimensional predictive modeling of zero-inflated count phenotypes to identify genetic susceptibility markers in Rheumatoid Arthritis patients	
Intern, Biostatistics Novartis Pharmaceuticals Corporation	Summer 2014
Developed novel Bayesian methods to conduct heterogeneity of treatment effect (HTE) analyses in phase III clinical trials	
Intern, Biostatistics University of Arkansas for Medical Sciences	Summer 2013
Developed methods for detecting maternal-fetal gene-gene interactions associated with obstructive heart defects in newborns from mother-offspring paired genetic data	
Intern, Biostatistics Indian Statistical Institute	Summer 2008
Performed research on non-linear statistical modeling of cross-sectional growth curve data by the Preece-Baines growth model	

COMPUTER SKILLS	Programming Languages: C, C++, Python, Julia, SQL Deep Learning: Keras, PyTorch, TensorFlow Statistical Software: SAS, R, MATLAB, SPSS Computing Platforms: Windows, Macintosh, Linux Applications: MS Office, L ^A T _E X, Adobe Illustrator, Shiny, Git
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NOTABLE SOFTWARE	MaAsLin2 MelonnPan Tweedieverse BayesRecipe IntegratedLearner
TOP 5 FIRST-AUTHOR PUBLICATIONS (TOTAL = 60+, CITATIONS = 13040, H-INDEX = 26, I-10 INDEX = 38)	Mallick et al. (2024). An Integrated Bayesian Framework for Multi-omics Prediction and Classification. <i>Statistics in Medicine</i> 43(5):983–1002. [Citations: 41] Mallick et al. (2022). Differential Expression of Single-cell RNA-seq Data using Tweedie Models. <i>Statistics in Medicine</i> 41(18):3492–3510. [Citations: 48] Mallick et al. (2021). Multivariable Association Discovery in Population-scale Meta-omics Studies. <i>PLoS Computational Biology</i> 17(11):e1009442. [Citations: 3,069] Mallick et al. (2019). Predictive Metabolomic Profiling of Microbial Communities Using Amplicon or Metagenomic Sequences. <i>Nature Communications</i> 10(1):3136–3146. [Citations: 334] Mallick and Yi (2014). A New Bayesian LASSO. <i>Statistics and Its Interface</i> 7(4):571–581. [Citations: 90]

SELECT HONORS	2026 Ritu Banga Impact in Healthcare Research Award, Weill Cornell Medicine 2024 Rising Star Award, UAB National Alumni Society 2023 Elected Member, International Statistical Institute 2023 Fellow, American Statistical Association (ASA) 2022 Early Career Award in Statistics and Data Sciences (ECASDS), IISA 2018 IMS New Researcher Travel Award , IMS 2017 Best Paper Award , Science Unbound Foundation 2016 AMS-Simons Travel Grant , American Mathematical Society 2015 Best Student Paper Award , American Statistical Association 2014 Best Student Paper Award , International Indian Statistical Association 2014 Grants-in-aid of Research , Sigma Xi 2013 Naomi Fineberg Distinguished New Researcher Award , UAB 2012 Best Talk , UAB Graduate Students Research Day
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